

Automated Protein Motif Localization using Concept Activation Vectors in Protein Language Model Embedding Space

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Abstract

We present an automated approach for identifying and annotating motifs and domains in protein sequences, using pretrained Protein Language Models (PLMs) and Concept Activation Vectors (CAVs), adapted from interpretability research in computer vision. We treat motifs as conceptual entities and represent them through learned CAVs in PLM embedding space by training simple linear classifiers to distinguish motif-containing from non-motif sequences. To identify motif occurrences, we extract embeddings for overlapping sequence windows and compute their inner products with motif CAVs. This scoring mechanism quantifies how strongly each sequence region expresses the motif concept and naturally detects multiple instances of the same motif within the same protein. Using a dataset of sixty-nine well-characterized motifs with curated positive and negative examples, our method achieves over 85% F1 Score for segments strongly expressing the concept and accurately localizes motif positions across diverse protein families. As each motif is encoded by a single vector, motif detection requires only the pretrained PLM and a lightweight dictionary of CAVs, offering a scalable, interpretable, and computationally efficient framework for automated sequence annotation.

1. Introduction

Structural motifs and domains represent the fundamental functional units of proteins, essentially modular units that can be shuffled, duplicated, and combined across different proteins to create diverse biological functions. These conserved sequence regions, ranging from short linear motifs of 3-10 amino acids to larger domains of 50-250 residues, encode specific molecular functions from catalytic sites and binding interfaces to regulatory elements and structural scaffolds. Accurately identifying the presence or absence of motifs in proteins is essential to understand protein function. Traditional annotation pipelines including Pfam, InterPro, and PROSITE have successfully catalogued thousands of motifs using sequence alignments, profile Hidden Markov Models (HMMs), and regular expressions [2, 6, 17]. Tools such as HMMER and FIMO enable scanning for new occurrences of known motifs using profile HMMs or position-specific scoring matrices [4, 7]. While these approaches have been invaluable, their dependence on explicit sequence similarity and curated motif patterns limits their ability to detect conserved motifs with divergent sequences, remote homologs, or novel variants that maintain function despite sequence variation.

Protein language models offer the opportunity for fundamentally different approaches to motif detection. PLMs such as ESM-C [5] are trained on millions of protein sequences, and produce distributed representations that capture biochemical and structural patterns. Essentially, PLMs represent amino acids in a sequence as high-dimensional embeddings that reflect their role in the

broader sequence context. Crucially, these learned representations can reflect functional and structural similarity between proteins even when sequence identity is low, as the model captures abstract patterns rather than explicit amino acid matches [13, 15].

Several recent studies have explored both PLMs and other transformer models for motif and domain identification. Here, we highlight four fundamentally different approaches. MotifAE [9] trains sparse autoencoders on PLM representations, identifying which learned features correspond to known functional motifs. The Encyclopedia of Domains [11] determines consensus domains from the AlphaFold database of protein structures, identifying more than 100 million novel domains. Chat-based models such as Evolla [22] are trained to connect protein sequences directly to natural language functional descriptions, allowing users to query for protein domain presence conversationally. Finally, the InterPro-N (previously PFAM-N) model [16] applies a transformer architecture inspired by panoptic segmentation to assign residue-level domain labels and distinguish repeated domain instances. These approaches demonstrate the breadth of current efforts to improve motif and domain annotation, but they generally rely on specialized architectures or large-scale training pipelines. In contrast, our method takes a conceptually orthogonal perspective, focusing on how motifs can be represented directly as directions in pretrained embedding space.

Here, we introduce a novel approach in which each motif or domain is treated as a distinct *concept* encoded as a direction in the PLM representation space. We adapt Concept Activation Vectors (CAVs), an interpretability framework originally developed for understanding neural networks [10], to protein sequence analysis. CAVs identify directions in the embedding space that correspond to human identifiable concepts. In the original framework, linear classifiers are trained to distinguish examples with and without a concept, such as images with and without stripes, and the vector perpendicular to the resulting decision boundary (the classifier’s weight vector) serves as a CAV for that concept. The alignment between these concept vectors and a query, as measured by inner product, quantifies how strongly the query expresses the concept. For instance, images of zebras align strongly with the concept of stripes.

We apply this principle to protein motifs: by training linear classifiers to distinguish motif from non-motif sequences, we obtain CAVs that capture the concepts of protein motifs. The inner product between embeddings of sequence windows (or regions) and a motif’s CAV quantifies how strongly each position expresses the corresponding biochemical function, producing alignment scores that peak at motif locations. Multiple peaks naturally indicate multiple motif instances within the same protein.

A key advantage of this approach to motif detection is its simplicity. Each motif is represented by a single vector in embedding space, and motif detection becomes a simple inner product computation between these vectors and sequence window embeddings. Notably, once a CAV is obtained for a motif, the linear classifier itself can be discarded, as motif detection for a protein requires only an embedding produced by the pretrained PLM and one stored vector per motif. This makes the method highly interpretable, computationally efficient, and trivially extensible to new motifs.

Our objective is to demonstrate that such concept-based geometric representations can be used to localize motifs in a scalable and interpretable manner. Specifically, we show that linear classifiers trained on pretrained PLM embeddings can produce motif-specific CAVs, enabling accurate detection of known motifs directly from representation space. We evaluate this framework across a curated set of well-characterized motifs and analyze layer-wise behavior of the underlying embeddings. Our results show that this simple approach achieves 90% precision and 80% recall in motif localization, demonstrating that functional motifs are indeed encoded as interpretable linear directions in PLM representation space.

2. Methodology

Our approach consists of two major stages: (i) learning motif-specific Concept Activation Vectors (CAVs), and (ii) applying these vectors to perform localized motif detection in unannotated sequences. Each step is designed to preserve biological interpretability while taking advantage of the representational strength of pretrained protein language models. Figure 1 shows an overview of the entire pipeline.

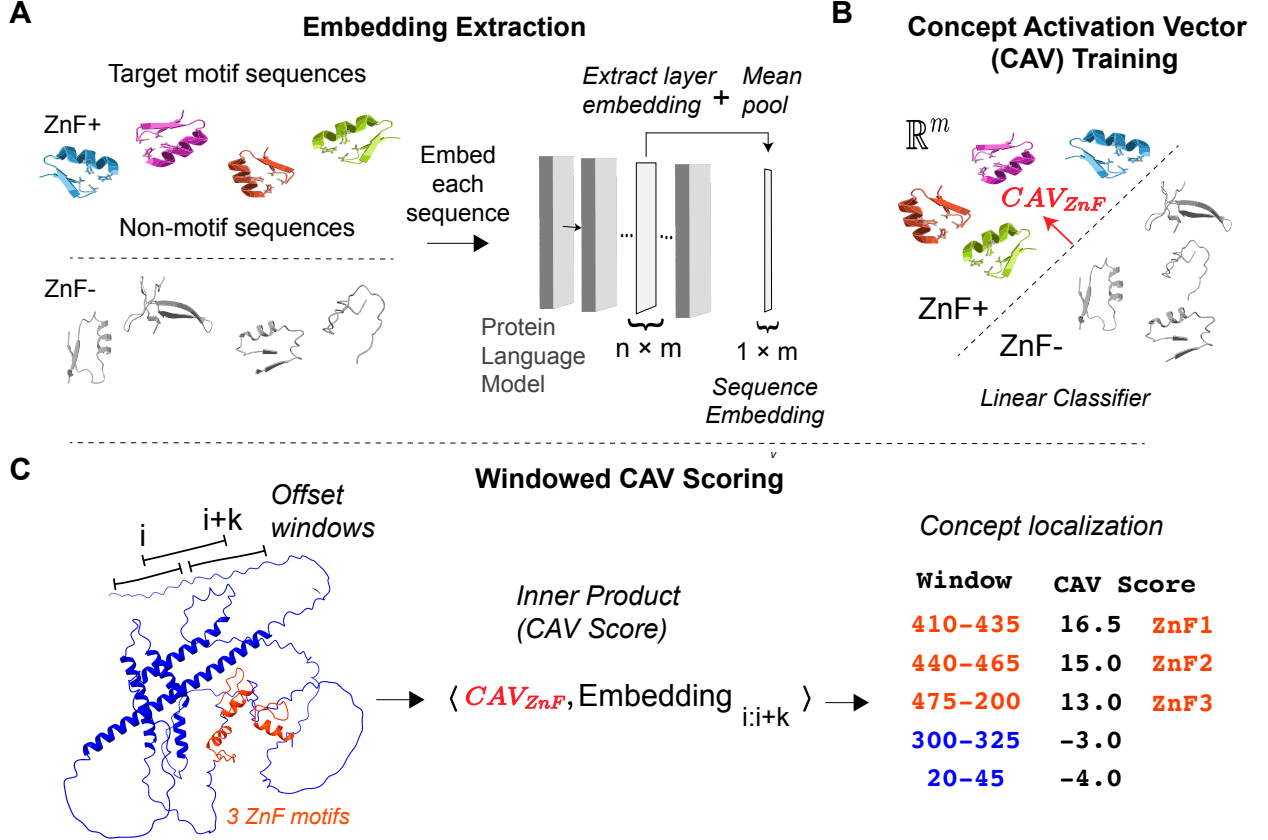


Figure 1: Overview of the proposed motif localization pipeline. (A) We first extract pooled embeddings from protein subsequences with and without a given motif (e.g., ZnF). These embeddings are ideally taken from a mid-late layer of the PLM. (B) These positive and negative examples are then used to train a linear classifier, with the vector orthogonal to the boundary defining the Concept Activation Vector (CAV) for that motif. (C) Finally, for an unknown protein, candidate segments are aligned with the CAV via inner products, where higher scores indicate stronger evidence of the motif.

Stage I: Learning Motif Concept Vectors

Constructing Windowed Training Samples. For each Pfam motif, we first construct a training dataset consisting of short subsequences (“windows”) drawn from proteins known to contain that motif (Figure 1A). Positive windows are extracted by centering a window of length w on each annotated motif instance, where w is defined as the median annotated motif length plus a small buffer to accommodate natural variation across homologs. We also generate negative windows of